

altairsim — Sol-20 cassette games

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A Processor Technology **Sol-20** with a 1977 game in the cassette deck. Four of them, each in its own machine file — pick one and run it:

```
altairsim trek80.toml      # TREK80  -- Star Trek
altairsim atc.toml        # ATC      -- Air Traffic Controller
altairsim pacman.toml     # PAC-MAN
altairsim raiders.toml    # RAIDERS
```

The game loads itself. Each file mounts the cassette, types the SOLOS command that runs it, and boots — so after the tape finishes loading (about the minute a real cassette took, at the Sol's own speed) the game comes up. TREK80 asks for a speed factor and starts; the others run straight away.

You do not have to type anything — but on a real Sol-20 you would, and the machine files simply do it for you. At the SOLOS > prompt you would enter `XE` and the file's five-character name; a `TYPE` line in each `startup` queues exactly that, and SOLOS reads it as type-ahead at its first prompt, as if the keys had been pressed. The names, for when you `XE` one yourself after a `REW`:

Game	Command
TREK80	<code>XE TRK80</code>
ATC	<code>XE ATC</code>
PAC-MAN	<code>XE PACMA</code> — SOLOS names are five characters
RAIDERS	<code>XE RAID</code>

What you are looking at

With SDL3 the screen is a **window**, in the VDM-1's own character font — the Sol-20's display was memory-mapped video at `0CC00H`, and that is what the games paint on. Headless, the same machine runs identically and the screen is readable with `DUMP CC00`, which is how the acceptance test checks the example.

`^E` (ATTN) takes the keyboard back to the monitor at any point; `RUN` resumes. The machine is not disturbed by stopping it.

The clock is 2.045 MHz, the stock Sol-20 — the 14.31818 MHz dot clock divided by 7 (Sol Systems Manual, Theory of Operation §VIII). Everywhere else in this simulator the default is flat out; here the real speed is set on purpose, because these are games and they were played at that speed. It is also why the tape takes a real minute to load. `SET cpu0 clock_hz=0` puts it back to flat out.

To play a game a second time, rewind first — nothing inside the guest can wind a tape back:

```
REW sol0:tape1
XE TRK80
```

The files

File	What it is
<code>trek80.toml</code> · <code>atc.toml</code> · <code>pacman.toml</code> · <code>raiders.toml</code>	The four machines: <code>base = "sol20"</code> , a tape, the launch keystrokes, and the clock.
<code>TRK80.WAV</code> · <code>ATC.WAV</code> · <code>PACMAN.WAV</code> · <code>RAIDERS.WAV</code>	The tapes. Sol-20 cassettes digitized by Philip Lord (hosted on deramp.com) — real CUTS audio at 1200 baud, decoded the way the hardware did.
<code>TRK80.TAP</code>	TREK80 as a byte stream, decoded from <code>TRK80.WAV</code> . <code>MOUNT sol0:tape1 "TRK80.TAP"</code> skips the audio and loads the same game.
<code>Trek80 Manual.pdf</code> · <code>ATC Manual.pdf</code>	The game manuals — commands, displays, and scoring — from Processor Technology and Creative Computing.
<code>TREK80.ENT</code>	A SOLOS <code>ENTER</code> script from the archive. The source for the tape-writing demonstration below.
<code>make-trek80-tape.sh</code>	A demonstration: it has SOLOS write its own tape from <code>TREK80.ENT</code> .

The tapes are Philip Lord's real recordings, and reading them is the point. Each `.WAV` decodes to a valid SOLOS tape with **zero framing errors** — for TREK80 the header names `TRK80` , its `SIZE` is `1EA0` , and its checksum is `D9` , all matching the archived `ENTER` script byte for byte. That did not used to work: an earlier decoder garbled these exact recordings (dozens of framing errors, most of the payload wrong), and the TREK80 example tape had to be *synthesized* from the `.ENT` instead. The demodulator now reads the real cassettes, so the real cassettes are what ship.

The synthesis is kept as `make-trek80-tape.sh` because it demonstrates the one thing a SOLOS tape cannot fake: a SOLOS cassette carries a header checksum and one after every 256-byte block, and a tape whose checksums are wrong is *invisible* — `CA` lists nothing and `GE` never finds the file. The script has SOLOS `SA` ve the image so the checksums are the machine's own arithmetic; it writes to a separate `TRK80-solos.*` so it never overwrites the shipped recording,

and it proves the two agree — SOLOS computes the same `D9` header checksum the real tape carries.

See `docs/sources.md` for full provenance.